

CS471 – Lesson 7 In-Class Worksheet

Heuristics, Admissibility, and Consistency

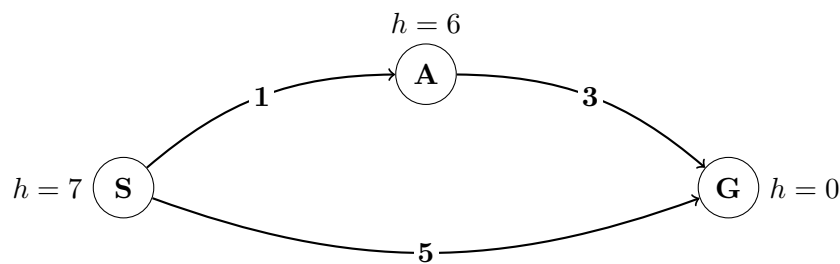
Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Search Algorithms Venn Diagram

Draw a Venn diagram showing the relationship between the following searches: **DFS**, **BFS**, **UCS**, **Greedy**, **A***. Organize them by cost (forward looking, backward looking, both, and none). Note which ones are considered “informed” searches.

2. Why A* Breaks

Evaluate the graph below. The true optimal solution is $S \rightarrow A \rightarrow G$ with a cost of 4. However, A* breaks on this graph and will select the suboptimal path $S \rightarrow G$ (cost 5).



Why does A* break here? What would the heuristic at node A need to be to fix this?

3. Core Definitions

Define the following foundational heuristic properties:

1. **Admissibility:**

2. **Consistency:**

3. **Dominance:**

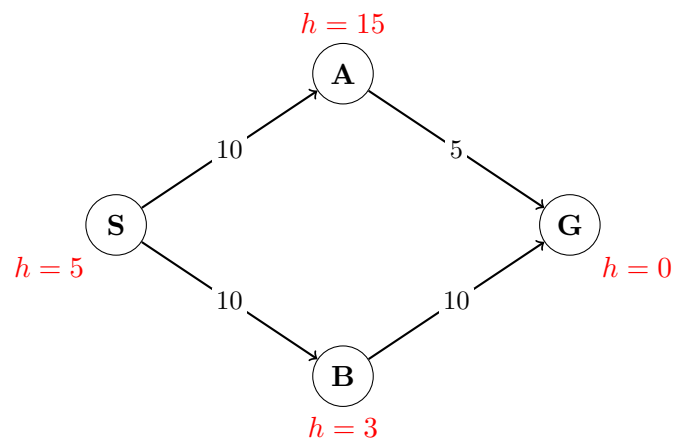
Part 2: Student Independent Practice

Objective: Independently evaluate search graphs for heuristic property violations and analyze combinations of admissible heuristics.

4. Evaluating Heuristic Properties

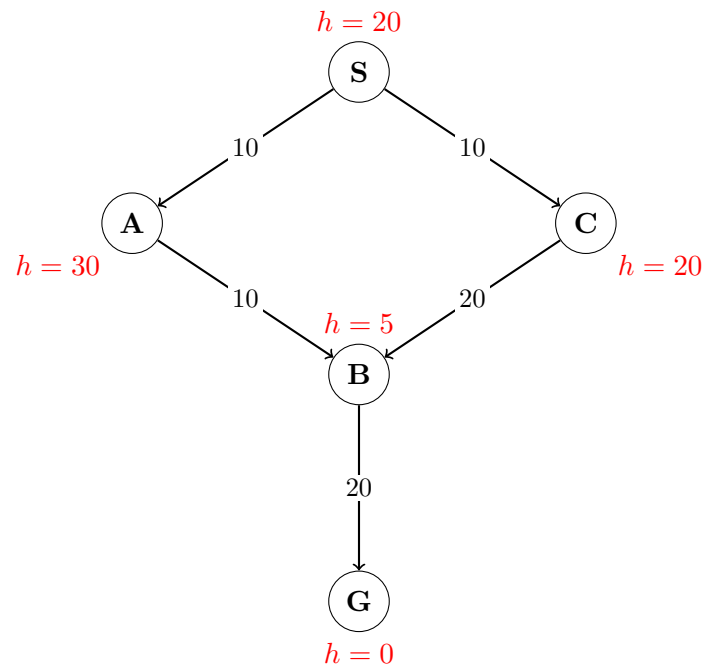
For each of the two graphs below, determine what property is violated (*none*, *admissibility only*, *consistency only*, or *both*). Assume the step costs are the black numbers on the arrows, and the heuristic estimates (h) are written in red next to the nodes.

Graph 1:



Violations & Proof:

Graph 2:



Violations & Proof:

5. Combining Admissible Heuristics

If $a(n)$, $b(n)$, and $c(n)$ are all **admissible** heuristics, is each of the following combined heuristics guaranteed to be admissible? (Yes/No, and explain why).

1. $a(n) + b(n) + c(n)$
2. $\min\{a(n), b(n), c(n)\}$
3. $\max\{a(n), b(n), c(n)\}$
4. $\frac{a(n)+b(n)+c(n)}{3}$
5. 0